

Executive Summary

Employee attrition represents a significant operational and financial challenge for technology organizations, particularly those dependent on specialized skills and technical expertise. High turnover can lead to increased recruitment costs, productivity disruptions, loss of organizational knowledge, and reduced workforce stability.

This project was undertaken to analyze workforce attrition patterns within Palo Alto Networks and identify the employee groups, departments, and workplace factors most strongly associated with employee turnover. The objective was to provide HR leadership with actionable insights that support evidence-based retention strategies and workforce planning decisions.

Using employee demographic, departmental, tenure, and workload-related data, the analysis evaluated attrition trends across multiple dimensions, including department, job role, age group, career stage, overtime status, business travel frequency, and commuting distance.

The organization recorded an overall attrition rate of 16.12%, indicating that approximately one in every six employees exited the company during the period analyzed. Department-level analysis identified Sales as the highest-risk department, followed by Human Resources, while Research & Development demonstrated comparatively stronger retention performance.

Role-based analysis revealed that Laboratory Technicians, Sales Executives, and Research Scientists experienced the highest exit frequencies, indicating elevated turnover among operational and frontline workforce segments. Leadership and senior management roles exhibited substantially lower attrition levels.

Demographic analysis showed that younger employees represented the most vulnerable workforce segment. Employees aged 18–25 experienced attrition rates significantly above the organizational average, suggesting that early-career professionals are more likely to seek external opportunities or career progression elsewhere. Single employees also demonstrated higher turnover compared to married and divorced employees.

Tenure analysis revealed that attrition was heavily concentrated among employees within their first two years at the company. Attrition declined steadily as employee tenure increased, indicating that workforce stability improves substantially after the initial employment period.

Among all factors examined, overtime emerged as the strongest indicator of attrition risk. Employees working overtime exhibited attrition rates nearly three times higher than employees who did not work overtime. Similarly, employees who travelled frequently for business demonstrated considerably higher turnover than those with limited or no travel requirements. Attrition also increased with commuting distance, highlighting the potential impact of travel burden on employee retention.

Based on these findings, the project recommends prioritizing early-career retention initiatives, reducing excessive overtime workloads, strengthening onboarding and mentorship programs, optimizing travel requirements, and implementing targeted interventions within high-risk departments and job roles.

The analysis provides a comprehensive view of workforce attrition patterns and establishes a data-driven foundation for strategic HR decision-making. By focusing retention efforts on the employee groups identified as most vulnerable, Palo Alto Networks can improve workforce stability, reduce replacement costs, and strengthen long-term organizational performance.